

GYAN GANGA INSTITUTE OF TECHNOLOGY & SCIENCES

Jabalpur, Madhya Pradesh | AICTE Approved | Affiliated to RGPV Bhopal

B.Tech — Artificial Intelligence & Machine Learning

COMPLETE SYLLABUS GUIDE

Semester 1 to Semester 8 | RGPV Curriculum 2024

■ Seats: 120	■ Fee: ■70,000/Year	■ Duration: 4 Years	■ Jabalpur, MP	■ NBA Accredited
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ABOUT GGITS — GYAN GANGA INSTITUTE OF TECHNOLOGY & SCIENCES

■ Established	2003	■ Affiliation	RGPV, Bhopal
■ Location	Tilwara Ghat, Near Bargi Hills, Jabalpur 482003	■ Approval	AICTE + PCI + NBA Accredited
■ Placement %	96% (2026)	■ Highest Package	■65 LPA (Texas Instruments)
■ Total Offers	1500+ Job Offers (2026)	■ NEC Ranking	Rank 3 PAN INDIA
■ Top Recruiters	TCS, HCL, Wipro, Infosys, Texas Instruments, Amazon	■ Library	9500+ Titles 59,647 Volumes

ABOUT B.Tech — AI & MACHINE LEARNING (AIML)

The B.Tech in Artificial Intelligence & Machine Learning (AIML) at GGITS is one of the most sought-after programs in Central India. Designed to meet industry demands, this 4-year program prepares students in cutting-edge technologies including Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Science, and Robotics. Affiliated with RGPV Bhopal and approved by AICTE.

Total Seats	Annual Fee	Total Fee (4 Yr)	Eligibility	Admission Basis
120	■70,000	■2,80,000	10+2 PCM 50%+	JEE Main / MP Counselling

SEMESTER 1	Subject Name	Credits	Type
BT-101	Engineering Mathematics - I	4	Theory

BT-102	Engineering Physics	4	Theory
BT-103	Basic Electrical & Electronics Engineering	4	Theory
BT-104	Engineering Mechanics	4	Theory
BT-105	Communication Skills	3	Theory
BT-106	Engineering Physics Lab	1	Practical
BT-107	Basic EEE Lab	1	Practical
BT-108	Workshop / Engineering Graphics	2	Practical
BT-109	Communication Skills Lab	1	Practical

SEMESTER 2	Subject Name	Credits	Type
BT-201	Engineering Mathematics - II	4	Theory
BT-202	Engineering Chemistry	4	Theory
BT-203	Introduction to Programming (C Language)	4	Theory
BT-204	Engineering Drawing & Graphics	3	Theory
BT-205	Environmental Studies	3	Theory
BT-206	Engineering Chemistry Lab	1	Practical
BT-207	Programming Lab (C)	2	Practical
BT-208	Engineering Drawing Lab	1	Practical

SEMESTER 3	Subject Name	Credits	Type
AL-301	Technical Communication	3	Theory
AL-302	Introduction to Probability & Statistics	4	Theory
AL-303	Data Structures & Algorithms	4	Theory
AL-304	Artificial Intelligence	4	Theory
AL-305	Object Oriented Programming (Python/Java)	4	Theory
AL-306	Technical Communication Lab	1	Practical
AL-307	Data Structures Lab	2	Practical
AL-308	OOP Lab (Python/Java)	2	Practical
AL-309	AI Lab	1	Practical

SEMESTER 4	Subject Name	Credits	Type
AL-401	Introduction to Discrete Structure & Linear Algebra	4	Theory
AL-402	Analysis & Design of Algorithms	4	Theory
AL-403	Software Engineering	4	Theory
AL-404	Computer Organization & Architecture	4	Theory

AL-405	Machine Learning	4	Theory
ES-401	Energy & Environmental Engineering	3	Theory
AL-406	Machine Learning Lab	2	Practical
AL-407	Algorithm Design Lab	1	Practical
AL-408	Mini Project - I	2	Practical

SEMESTER 5	Subject Name	Credits	Type
AL-501	Operating Systems	4	Theory
AL-502	Database Management Systems	4	Theory
AL-503A	Deep Learning (Elective)	4	Elective
AL-503B	Information Retrieval (Elective)	4	Elective
AL-504A	Natural Language Processing (Elective)	4	Elective
AL-504B	AI in Healthcare (Elective)	4	Elective
AL-505	Deep Learning / NLP Lab	2	Practical
AL-506	Operating Systems Lab	1	Practical
AL-507	DBMS Lab	1	Practical

SEMESTER 6	Subject Name	Credits	Type
AL-601	Theory of Computation	4	Theory
AL-602	Computer Networks	4	Theory
AL-603A	Data & Visual Analytics (Elective)	4	Elective
AL-603B	Big Data Analytics (Elective)	4	Elective
AL-604A	Cloud Computing (Elective)	4	Elective
AL-604C	Intelligent Systems for Robotics (Elective)	4	Elective
AL-605	Computer Networks Lab	2	Practical
AL-606	Cloud / Analytics Lab	2	Practical
AL-607	Mini Project - II	2	Practical

SEMESTER 7	Subject Name	Credits	Type
AL-701	Computer Vision	4	Theory
AL-702A	AI in Gaming (Elective)	4	Elective
AL-702B	Advanced Machine Learning (Elective)	4	Elective
AL-702C	Predictive Analytics (Elective)	4	Elective
AL-702D	Machine Learning for Data Science (Elective)	4	Elective
AL-703B	Augmented & Virtual Reality (Elective)	4	Elective

AL-703C	Agile Software Development (Elective)	4	Elective
AL-703D	Internet & Web Technology (Elective)	4	Elective
AL-704	Computer Vision Lab	2	Practical
AL-705	Internship / Industrial Training	4	Practical

SEMESTER 8	Subject Name	Credits	Type
AL-801	Big Data	4	Theory
AL-802B	Reinforcement Learning (Elective)	4	Elective
AL-803	Managing Innovation & Entrepreneurship	4	Theory
AL-804	Project Work (Major Project)	8	Project
AL-805	Seminar & Technical Presentation	2	Seminar
AL-806	Comprehensive Viva Voce	2	Viva

CAREER PROSPECTS & PLACEMENT DATA (GGITS 2026)

Role	Company Examples	Avg Package
ML Engineer	Google, Microsoft, Amazon, Infosys	■8-20 LPA
Data Scientist	TCS, Wipro, HCL, Capgemini	■6-15 LPA
AI Research Analyst	IBM, Oracle, Adobe	■7-18 LPA
Software Engineer (AI)	Texas Instruments, Qualcomm, Samsung	■5-65 LPA
GGITS Highest Package	Texas Instruments	■65 LPA
GGITS Average Package	85+ Companies visited campus in 2026	■3.34 LPA

GGITS FACILITIES & ACHIEVEMENTS

■ Campus	10-acre campus Wi-Fi enabled Modern classrooms AC Labs
■ Library	9500+ Titles 59,647 volumes E-books Digital resources
■ Hostel	Separate boys & girls hostel 24/7 security Mess facility
■ Labs	HCL K2 Academy Cisco Networking Academy Microsoft Innovation Center
■ Placements	96% placement (2026) 85+ companies 1500+ offers Texas Instruments highest ■65 LPA
■ Rankings	NIRF Innovation: Rank 151-300 NEC PAN INDIA Rank 3 Careers360: National Rank A
■ Scholarships	AICTE Pragati AICTE Saksham MP Govt SC/ST/OBC Minority Merit-based
■ Research	NEETI AYOOG ■2.5Cr Incubation Centre IETE IEEE Computer Society chapter

